Bayesian Inference and Sensitivity Analysis of a Jansen–Rit Neural Mass Model

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1 Introduction

In seeking comprehensive understanding of the brain, there are applications for which forward modeling does not provide sufficient insight. Inferring brain states from EEG data requires inverse modeling to estimate underlying parameters from observations. This approach introduces challenges such as noise, signal aggregation, and structural non-identifiability [Raue et al., 2009], which hinder recovery of true parameter values. In high-dimensional models such as Jansen–Rit, identifying parameters that most influence observable output is critical for inference.

Sensitivity analysis and parameter inference both are critical in computational neuroscience, as they increase the interpretability of neural signals and model behavior. Both involve inherent uncertainty, which must be quantified to understand their limitations. In this study, the Jansen–Rit model is used to perform sensitivity analysis and Bayesian parameter inference via the Metropolis algorithm. The analysis characterizes the sensitivity and identifiability of two key parameters, and discusses how these qualities affect model predictions.

2 Jansen–Rit Implementation

The Jansen–Rit cortical column model allows high dimensional retention of biophysical detail that other models, such as the Wilson-Cowan model, abstract away [Jansen and Rit, 1995, Wilson and Cowan, 1972]. Its nonlinear dynamics make it well-suited for sensitivity analysis and parameter inference, as relationships between parameters and observables are not immediately apparent. The model supports multiple dynamical regimes reflected in the pyramidal cell input, an EEG-like proxy signal [Buzsáki et al., 2012], facilitating qualitative evaluation of parameter sensitivity. The equations used in this implementation follow Jansen and Rit [1995].

$$
\begin{aligned}
\dot{y}_0(t) &= y_3(t), \\
\dot{y}_1(t) &= y_4(t), \\
\dot{y}_2(t) &= y_5(t), \\
\dot{y}_3(t) &= Aa\, S(y_1(t) - y_2(t)) - 2a\, y_3(t) - a^2 y_0(t), \\
\dot{y}_4(t) &= Aa\, [p(t) + C_2\, S(C_1 y_0(t))] - 2a\, y_4(t) - a^2 y_1(t), \\
\dot{y}_5(t) &= Bb\, C_4\, S(C_3 y_0(t)) - 2b\, y_5(t) - b^2 y_2(t).
\end{aligned}
\tag{1}
$$

$$
S(v) = \frac{2e_0}{1 + \exp\big(r(v_0 - v)\big)} \tag{2}
$$

2.1 Parameters to be Inferred

This study focuses on parameters A and B, the excitatory and inhibitory gain constants that scale population activity. A scales the activity of pyramidal and excitatory interneuron populations, while B scales inhibitory interneuron activity. Sensitivity analysis examines how small changes in A and B affect the observable output $y_1 - y_2$. Bayesian inference using the Metropolis algorithm is then applied to assess parameter identifiability.

3 Synthetic Data Generation

To generate the observed data for the inference study, a simulation of the Jansen–Rit model was first run using the ground truth parameter values of $A = 3.25$ and $B = 22.0$. An IVP solver was used to produce the solution of the six differential equations. The clean data produced from this run was then altered with additive Gaussian noise to emulate variability. Noise was drawn from a zero-mean normal distribution with a standard deviation $\sigma = 0.15$, and a fixed random seed was used to ensure data reproducibility. This noise level is on the order of the signal amplitude, preserving waveform structure while preventing perfect recovery. This reflects realistic EEG inference conditions, where measurements are noisy and parameter estimation is uncertain. A visualization of the synthetic data can be generated via the project files but is omitted here for brevity.

4 Sensitivity Analysis

4.1 Method

A qualitative parameter sweep was conducted to build intutition for parameter sensitivity. To conduct this sweep, simulations varied A by ± 0.25 around 3.25 and B by ± 2.0 around 22.0. Results were compared to the baseline signal to assess how each parameter affects the output.

$$S_t = \frac{y_{pos} - y_{neg}}{2\delta} \tag{3}$$

Local sensitivity was then computed using central finite differences around the ground-truth parameters to obtain a more rigorous understanding of parameter sensitivity [O'Reilly, 2026b]. Step sizes of $\delta_A = 0.1$ and $\delta_B = 0.5$ were used. Sensitivity is computed as the approximate derivative of the output with respect to each parameter at each time point (Eq. 3). Sensitivity norms are computed using the root mean square, representing the average magnitude of sensitivity over the time series.

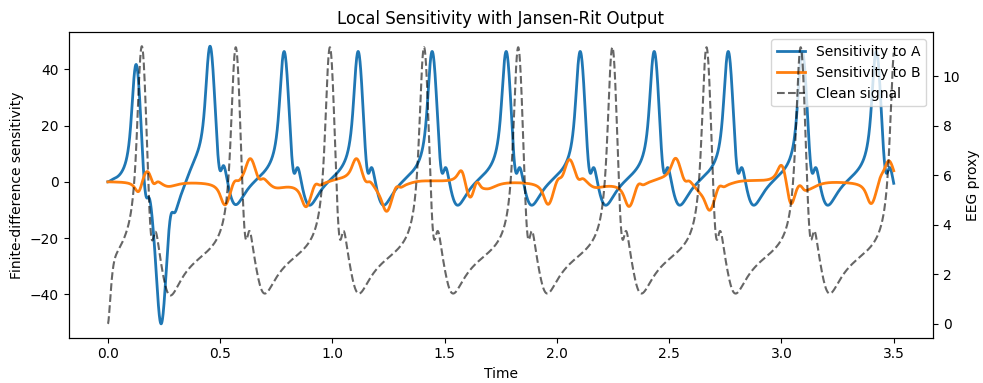

Figure 1: Results of Finite Difference Sensitivity Analysis

4.2 Results, Interpretation & Implications for Inference

The qualitative parameter sweep plot can be generated via the project files, but is omitted from the report for brevity. The sweep showed that increasing excitatory gain increases the amplitude of oscillatory spikes. Decreasing it appears to drive the system toward a fixed point. Increasing inhibitory gain produces an initial spike followed by convergence to equilibrium, indicating that inhibition dominates after the transient. Decreasing B produces damped oscillations, indicating loss of a stable limit cycle.

Finite Difference analysis (Fig. 1) shows that the proxy signal is more sensitive to changes in A than in B. This is quantified by sensitivity norms of 17.2 for A and 3.4 for B, under a simulation time of 3.5 seconds. This suggests that A is more identifiable than B, as small changes in A produce larger output variations. The weaker local sensitivity of B may cause true parameter values to be more easily obscured by complex model dynamics and noise.

Furthermore, sensitivity in A spikes during steep inclines or declines in the proxy signal, while B shows similar but much more muted oscillatory behavior. This effect reflects the nonlinear dynamics of the system, under which small perturbations can have amplified effects during specific phase windows of a system's oscillation.

5 Bayesian Inference

5.1 Likelihood Function & Prior

The Metropolis algorithm is used to estimate posteriors for the parameters A and B. The likelihood function assumes that the observed data is generated by the model output corrupted by Gaussian

noise. Under this assumption, discrepancies between observed and simulated signals are modeled as independent, normally distributed errors. Defining the error as the difference between observed and simulated signals at each time point, the log-likelihood is expressed as a scaled negative mean squared error, proportional to the Gaussian likelihood. We use the mean instead of the sum to stabilize the scale of likelihood for the MCMC walker:

$$likelihood = -\frac{mean(error^2)}{2\sigma^2} \tag{4}$$

where σ is the standard deviation of the observation noise. The function will assign higher values to parameter sets that produce data closely matching the observed data.

A flat (non-informative) prior was assumed, so inference is driven primarily by the likelihood. This reflects the assumption that no prior information about the ground-truth parameters is available. Under this assumption, all admissible values of A and B are equally likely a priori.

5.2 MCMC Method

We use the Metropolis sampler, as the symmetric Gaussian proposal distribution makes the Metropolis–Hastings correction unnecessary [O'Reilly, 2026a]. The Markov chain was run for 8000 steps with initial parameters $A = 3.15$ and $B = 23.5$. The noise standard deviation was set to $\sigma = 0.15$, consistent with the synthetic data generation, and a burn-in period of 2000 samples was used. Step sizes of 0.02 for A and 0.08 for B were used, with a simulation time of 1.5 seconds. After burn-in, the chain reaches a stationary regime in which sampled parameters fluctuate predictably.

5.3 Results

Initial simulations showed the Metropolis sampler becoming trapped in local basins rather than reaching high-likelihood regions. To address this, a coarse likelihood scan (Fig. 4) was conducted to characterize the geometry of the parameter space. This revealed a large basin that trapped the sampler and informed improved initialization. While scans are less feasible in higher-dimensional settings, analogous approaches include running multiple chains or performing broader initial sweeps. The posterior distribution indicates a diagonal region of strong parameter correlation.

Convergence was assessed using trace plots and the stabilization of the log-likelihood. The burn-in period of the MCMC traces precedes stable fluctuations of the traces with no persistent drift, which qualitatively approximates convergence to a stationary distribution. The log-likelihood is the strongest indicator of convergence, as it increases sharply at first and then plateaus, signaling

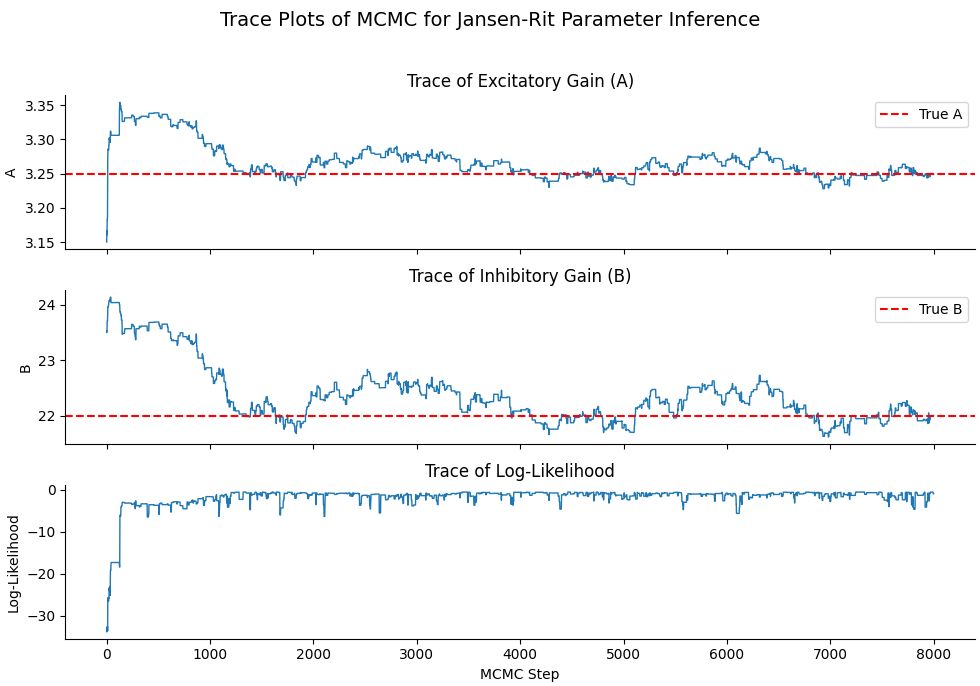

Figure 2: Trace plots for MCMC

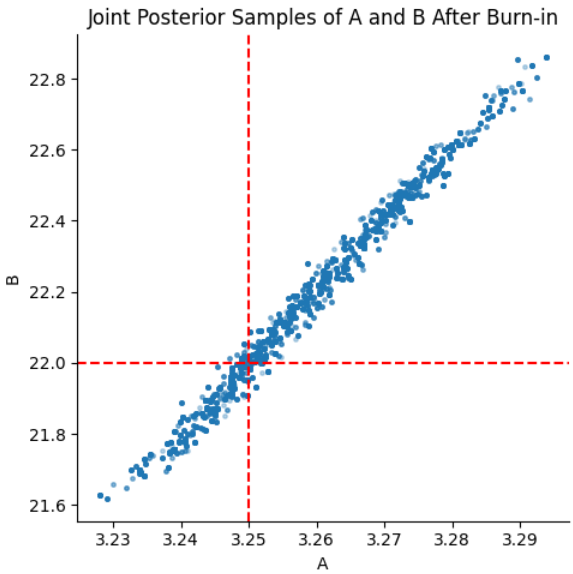

Figure 3: Joint posterior samples of (A, B) showing parameter correlation.

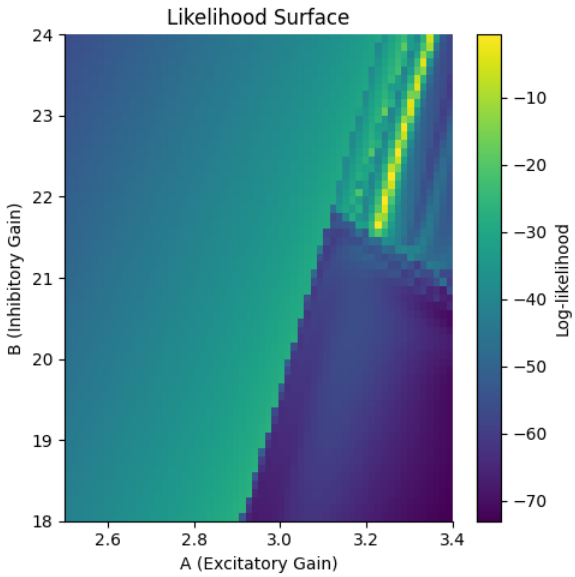

Figure 4: Likelihood surface over parameter space.

that the chain has reached a high-likelihood region. Quantitative measures can be used to assess convergence more rigorously.

6 Model Driven Analysis

The joint posterior in Fig. 3 exhibits a ridge in parameter space where log-likelihood is highest, indicating correlation between parameters. The likelihood surface and trace behavior indicate that the stationary distribution lies along this ridge rather than concentrating at a single point. This reflects partial non-identifiability, where multiple combinations of excitatory and inhibitory gains produce similar outputs. Accordingly, A and B cannot be reliably inferred independently, but balanced pairs can be inferred. The observed data constrains combinations of parameters but does not uniquely determine individual values. The orientation of the ridge suggests that model fit depends on the balance between A and B rather than their absolute values. This reflects the Jansen-Rit model structure, where excitation and inhibition interact in a compensatory manner.

The trace plots in Fig. 2 show the evolution of sampled parameters and provide a visual assessment of convergence and mixing. The initial transient shows rapid movement from the initial guess to a high-probability region before reaching a stationary distribution. The traces fluctuate near the true values rather than converging to a single point, reflecting posterior uncertainty. They also appear somewhat smooth due to autocorrelation between samples, as they are not independent. This arises from the use of local proposals in the Metropolis algorithm. Autocorrelation effectively reduces sample size, but is still valid for estimating posteriors given a long enough sampling horizon. Correlation between samples might be reduced with larger step sizes or reparameterization of the problem. The acceptance rate achieved here (about 8%) is lower than the typical target range (20–40%) and could be increased through similar adjustments [O'Reilly, 2026a].

Additionally, the stationary distribution forms a diagonal ridge of parameter combinations producing similar output (Fig. 3), and the trace plots reflect this through co-varying A and B values while log-likelihood remains stable, indicating movement along the ridge rather than independent variation. The lack of identifiability indicates that determining unique values for A and B without a prior is infeasible. Many combinations of the parameters produce similarly plausible outputs, meaning that the model doesn't yield a single deterministic prediction but rather a range of possibilities. These outputs may share features such as oscillation frequency while differing in features like amplitude, implying that some aspects of the predictions may be robust while others remain uncertain.

7 Conclusion

This study demonstrates how sensitivity analysis and parameter inference inform model behavior and output. Sensitivity analysis of excitatory and inhibitory gain of the Jansen-Rit model showed that model output is more strongly influenced by the former, particularly during periods of rapid change. Bayesian parameter inference revealed correlation between parameters and partial non-identifiability, indicating that similar model output can be reproduced by multiple parameter combinations, limiting unique parameter recovery. Potential improvements include reduction of autocorrelation to improve sampling efficiency, and exploring correlations with other parameters in the Jansen–Rit model to further assess identifiablity.

8 Implementation Details & Acknowledgments

The methods for this study were inspired by lecture material by Dr. Christian O'Reilly from CSCE790 – Introduction to Computational Neuroscience, taught at the University of South Carolina. This project used generative AI tools as development aids to help clarify concepts, troubleshoot, and refine code and structure. All equations, model behavior, parameter choices, etc. were implemented and validated by the author and represent the author's independent understanding.

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